

Chitralkha Gupta

MIT Media Lab

Dept. of Computer Science, National University of Singapore

Senior Research Fellow

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Research Profile

I develop human-centered, multimodal AI systems that support learning, critical thinking, and decision-making while preserving people's judgment and capacity to act independently. I aim to address learning and critical thinking problems faced by younger adults and ageing population. My research combines human-AI interaction, speech and audio processing, wearable sensing, and empirical evaluation to study when people need support, how much assistance to provide, and whether immediate benefits translate into lasting understanding and independent reasoning. My interests span adaptive voice and multimodal agents, wearable misinformation interventions, accessible scene perception, explainable speech assessment, and controllable generative audio.

Academic Appointments

Postdoctoral Researcher

MIT Media Lab, Fluid Interfaces Group

2026–Present

Cambridge, MA

Lead research on wearable systems that detect questionable spoken claims and deliver selective, evidence-oriented nudges during everyday activities for critical thinking.

Study preferences for control and autonomy in computer-use voice agents, and adaptive vocabulary learning with wearable AI.

Investigate multimodal agents that support learning and judgment while preserving user control.

Senior Research Fellow

Dept. of Computer Science, National University of Singapore

2023–Present

Singapore

Lead research across assistive scene sonification, explainable dysarthric-speech assessment, and drone audio perception.

Develop research proposals, mentor graduate students, and coordinate healthcare, accessibility, industry, and international collaborations.

Research Fellow

Dept. of Electrical and Computer Engineering, and Dept. of Communication and New Media, NUS

2019–2023

Singapore

Developed leading lyrics-alignment and transcription systems (MIREX 2019–2020), singing-quality assessment methods, and controllable audio generation.

Collaborated on human-centered applications of AI, including VR experiences and assistive audio technologies.

Entrepreneurship and Industry Experience

Co-Founder, MuSigPro Pte. Ltd.

Commercialized singing-quality assessment and lyrics-alignment technologies; reached more than 35,000 users. Translation of this research contributed to a U.S. patent.

2019–2025

Research Engineer, Airbus Defence & Space, Bangalore

2013–2014

Software Developer, Dell R&D, Bangalore

2011–2013

Education

Ph.D., Computer Science, National University of Singapore

Thesis: Comprehensive Evaluation of Singing Quality. CAP: 4.38/5.0.

Advisors: Haizhou Li and Ye Wang.

2015–2019

M.Tech., Electrical Engineering, IIT Bombay

Thesis: Objective Assessment of Ornaments in Indian Singing. GPA: 9.63/10.0.

Advisor: Preeti Rao.

2008–2011

B.E., Electronics Engineering, M.S. University, Baroda

2004–2008

Research Contributions

Wearable AI for critical thinking. Developed FactNudger and related wearable fact-checking systems to study how real-time nudges affect belief and verification. Extend this work through VibeCheck, investigating proactive support during everyday activities. (CHI 2026; UIST 2026 demo; CHI 2025 demo and workshop; CHI 2027 submission.)

Learning and human–AI agency. Investigate instructional sequencing in second-language vocabulary learning with wearable AI and how age and task characteristics shape preferences for computer-use voice-agent autonomy. (Two CHI 2027 submissions.)

Accessible scene perception. Developed SonicVista and Scene2Audio to translate distant visual scenes into nonverbal audio for blind and low-vision users, with controlled and in-the-wild evaluations. (IMWUT/UbiComp 2024; CHI-EA 2025; CHI 2026.)

Multimodal drone perception. Led the Singapore side of a Singapore–France research project on augmented human–drone interaction. Developed DroneAudioset and noise-suppression methods to support search and rescue beyond a camera’s field of view. (NeurIPS 2025; DroneAudioNet manuscript; CHI 2027 submission.)

Explainable AI for healthcare. Collaborated with Alexandra Hospital speech therapists on temporally localized, interpretable dysarthric-speech clarity assessment, supporting actionable pronunciation feedback. (Interspeech 2025.)

Generative audio and music intelligence. Developed semantic controls for audio generation and morphing, lyrics alignment and transcription, and reference-independent singing assessment. (IEEE/ACM TASLP; ICASSP; ISMIR; IUI; SMC.)

Research Funding and Leadership

Roles and total project allocations are listed separately; amounts are not personal funding.

Principal Investigator — AI for Audio Applications. NUS Enterprise, associated with the inaugural Innovation and Venture Creation Award. **SGD 160,000.** Awarded 2025.

Co-Principal Investigator — Context-aware decision support for human–robot interaction. NRF Translational Project supporting future Singapore Home Team operations. **SGD 1.47 million.** Awarded 2026.

Singapore Research Lead — Augmented Human-Swarm Interactions: Blind Drone. Singapore–France Ministry of Defence project; led ideation, proposal writing, and Singapore-side execution. **SGD 660,000** allocated to the Singapore team. Awarded 2024.

Collaborator — AI-based dysarthric-speech intelligibility assessment. MOE Tier 1/NUHS Seed Grant; contributed to the research proposal. **SGD 138,000.** Awarded 2024.

Collaborator — Augmented Awareness through Scene Sonification. NUS School of Computing Seed Grant; contributed to ideation and proposal development. **SGD 100,000.** Awarded 2023.

Research Collaborations

MIT Media Lab, USA	2025–Present
CNRS, Toulouse, France	2023–Present
Alexandra Hospital, Singapore	2024–2026
Meta Platforms Inc., USA	2022–2024
AIST, Japan	2021–2022

Teaching Experience

Co-teacher, AI Agents for Cognitive Augmentation (MAS.S60)

Fall 2026

Graduate course, MIT Media Lab [Course website](#)

Lead a lecture on wearable, multimodal, and physiology-aware agents: gaze, EEG, electrodermal activity, extended reality, audio, context sensing, and closed-loop intervention.

Lead a hands-on multimodal-agents tutorial integrating vision, audio, video, gaze, screen state, interaction traces, and sensor data.

Contributed to an experimental course-agent website for exploring agent concepts, examples, and design choices.

Guest Lecturer, National University of Singapore

2024–2025

IS5006 Human-Centred Intelligent Systems (2024–2025).

IS3251 Principles of Technology Entrepreneurship (2024).

Awards and Recognition

NUS Enterprise Innovation and Venture Creation Award, 2025: One of four inaugural recipients; associated PI grant for AI for audio applications.

IEEE VR Best Paper Award, 2024: VR.net.

DCASE Challenge, 2023: Third place in Foley Sound Synthesis among 26 submitted systems.

MIREX, 2019–2020: First place in lyrics alignment and transcription.

Graduate Research Innovation Program, 2019: Venture support for MuSigPro.

Dean's Graduate Research Achievement Award, 2018.

APSIPA ASC Best Student Paper Award, 2017: Perceptual Evaluation of Singing Quality.

Professional Service and Invited Talks

Area Chair (Analysis of Speech and Audio Signals): Interspeech 2026.

Outstanding review recognitions: One recognition each for UIST 2024 Papers and CHI 2025 Papers.

Conference organization: ISMIR 2022, ICASSP 2022, and ASRU 2019. Additional organizing committee service: ISMIR 2017, SIGDIAL 2021, and Augmented Human Workshop 2023.

SIGCHI reviewing: CHI Papers (2025, 2026, 2027), CHI Posters (2026), and UIST Papers (2024, 2026).

Additional reviewing: ICIS 2023, IEEE/ACM TASLP, IEEE Transactions on Multimedia, ISMIR, ICASSP, Interspeech, ICME, APSIPA Transactions, International Journal of Social Robotics, IEEE Access, and Multimedia Systems.

Invited talks: Meta ARIA Summit, Redmond (2024); IIT Bombay (2021). **Panelist:** Women in MIR, ISMIR (2022).

Patents

US Patent 11,972,774 (2024): System and Method for Assessing Quality of a Singing Voice. C. Gupta, H. Li, and Y. Wang.

Singapore application 10202403564W (2024): AI-generated sounds for scene understanding.

Singapore application 10202403845Y (2024): Fine-grained controllable audio morphing.

Published research is listed below, followed by preprints and manuscripts under review. An asterisk denotes equal contribution.

Journal Articles

1. **C. Gupta**, S. Sridhar, D. J. C. Matthies, C. Jouffrais, and S. Nanayakkara. *SonicVista: Towards Creating Awareness of Distant Scenes through Sonification*. ACM IMWUT/UbiComp, 2024.
2. F. F.-Y. Tan, **C. Gupta**, D. P. D. Rajendran, P. Maes, and S. Nanayakkara. *Assistive Augmentation: Fundamentally Transforming Human Ability*. ACM Interactions, 2025.
3. P. Kamath, **C. Gupta**, L. Wyse, and S. Nanayakkara. *Example-Based Framework for Perceptually Guided Audio Texture Generation*. IEEE/ACM TASLP, 2024.
4. E. Wen, **C. Gupta**, P. Sasikumar, M. Billingham, J. Wilmott, E. Skow, A. Dey, and S. Nanayakkara. *VR.net: A Real-world Dataset for Virtual Reality Motion Sickness Research*. IEEE TVCG / IEEE VR, 2024. **Best Paper Award**.
5. **C. Gupta***, S. Siriwardhana*, T. Kaluarachchi, V. Dissanayake, S. Ellawela, and S. Nanayakkara. *Can AI Models Summarize Your Diary Entries? Investigating Utility of Abstractive Summarization for Autobiographical Text*. International Journal of Human-Computer Interaction, 2024 (online 2023).
6. X. Gao, **C. Gupta**, and H. Li. *PoLyScriber: Integrated Fine-Tuning of Extractor and Lyrics Transcriber for Polyphonic Music*. IEEE/ACM TASLP, 2023.
7. **C. Gupta**, H. Li, and M. Goto. *Deep Learning Approaches in Topics of Singing Information Processing*. IEEE/ACM TASLP, 2022.
8. X. Gao, **C. Gupta**, and H. Li. *Automatic Lyrics Transcription of Polyphonic Music with Lyrics-Chords Multi-Task Learning*. IEEE/ACM TASLP, 2022.
9. **C. Gupta**, H. Li, and Y. Wang. *Automatic Leaderboard: Evaluation of Singing Quality without a Standard Reference*. IEEE/ACM TASLP, 2019.
10. **C. Gupta**, H. Li, and Y. Wang. *A Technical Framework for Automatic Perceptual Evaluation of Singing Quality*. APSIPA Transactions on Signal and Information Processing, 2018.

Conference Papers and Book Chapters

11. **C. Gupta**, N. V. Arintonang, D. P. D. Rajendran, V. Danry, P. Maes, and S. Nanayakkara. *Feeling the Facts: Real-time Wearable Fact-checkers Can Use Nudges to Reduce User Belief in False Information*. ACM CHI, 2026.
12. **C. Gupta**, J. Peng, A. Ram, S. Sridhar, C. Jouffrais, and S. Nanayakkara. *Beyond Descriptions: A Generative Scene2Audio Framework for Blind and Low-Vision Users to Experience Vista Landscapes*. ACM CHI, 2026.
13. **C. Gupta***, S. Ramesh*, P. Sasikumar, K. P. Yeo, and S. Nanayakkara. *DroneAudioset: An Audio Dataset for Drone-based Search and Rescue*. NeurIPS, Datasets and Benchmarks Track, 2025.
14. **C. Gupta***, S. Park*, M. K. Y. Kwan, X. Fung, A. W. Yip, and S. Nanayakkara. *Towards Temporally Explainable Dysarthric Speech Clarity Assessment*. Interspeech, 2025.
15. Y. Wei, N. Rocher, **C. Gupta**, M. H. Nguyen, R. Zimmermann, W. T. Ooi, C. Jouffrais, and S. Nanayakkara. *Human Robot Interaction for Blind and Low Vision People: A Systematic Literature Review*. ACM CHI, 2025.
16. P. Kamath, **C. Gupta**, and S. Nanayakkara. *MorphFader: Enabling Fine-grained Controllable Morphing with Text-to-Audio Models*. ICASSP, 2025.
17. M. H. Nguyen, Y. Samaradivakara, P. Sasikumar, **C. Gupta**, and S. Nanayakkara. *EMO-KNOW: A Large Scale Dataset on Emotion-Cause*. Findings of EMNLP, 2023.
18. **C. Gupta***, P. Kamath*, Y. Wei, Z. Li, S. Nanayakkara, and L. Wyse. *Towards Controllable Audio Texture Morphing*. ICASSP, 2023.
19. P. Kamath, Z. Li, **C. Gupta**, K. Jaidka, S. Nanayakkara, and L. Wyse. *Evaluating Descriptive Quality of AI-Generated Audio Using Image-Schemas*. ACM IUI, 2023.
20. **C. Gupta**, Y. Wei, Z. Gong, P. Kamath, Z. Li, and L. Wyse. *Parameter Sensitivity of Deep-Feature Based Evaluation Metrics for Audio Textures*. ISMIR, 2022.
21. X. Gao, **C. Gupta**, and H. Li. *Music-robust Automatic Lyrics Transcription of Polyphonic Music*. Sound and Music Computing (SMC), 2022.
22. X. Gao, **C. Gupta**, and H. Li. *Genre-conditioned Acoustic Models for Automatic Lyrics Transcription of Polyphonic Music*. ICASSP, 2022.

23. L. Wyse, P. Kamath, and **C. Gupta**. *Sound Model Factory: An Integrated System Architecture for Generative Audio Modelling*. EvoMUSART / Springer LNCS, 2022.
24. J. Li, **C. Gupta**, and H. Li. *Training Explainable Singing Quality Assessment Network with Augmented Data*. APSIPA ASC, 2021.
25. **C. Gupta**, J. Li, and H. Li. *Towards Reference-Independent Rhythm Assessment of Solo Singing*. APSIPA ASC, 2021.
26. **C. Gupta**, P. Kamath, and L. Wyse. *Signal Representations for Synthesizing Audio Textures with Generative Adversarial Networks*. SMC, 2021.
27. L. Huang, **C. Gupta**, and H. Li. *Spectral Features and Pitch Histogram for Automatic Singing Quality Evaluation with CRNN*. APSIPA ASC, 2020.
28. **C. Gupta**, L. Huang, and H. Li. *Automatic Rank Ordering of Singing Vocals with Twin-Neural Network*. ISMIR, 2020.
29. **C. Gupta**, E. Yilmaz, and H. Li. *Automatic Lyrics Alignment and Transcription in Polyphonic Music: Does Background Music Help?*. ICASSP, 2020.
30. **C. Gupta**, E. Yilmaz, and H. Li. *Acoustic Modeling for Automatic Lyrics-to-Audio Alignment*. Interspeech, 2019.
31. **C. Gupta***, B. Sharma*, H. Li, and Y. Wang. *Automatic Lyrics-to-Audio Alignment on Polyphonic Music Using Singing-Adapted Acoustic Models*. ICASSP, 2019.
32. **C. Gupta**, H. Li, and Y. Wang. *Automatic Evaluation of Singing Quality without a Reference*. APSIPA ASC, 2018.
33. **C. Gupta**, H. Li, and Y. Wang. *Automatic Pronunciation Evaluation of Singing*. Interspeech, 2018.
34. **C. Gupta**, R. Tong, H. Li, and Y. Wang. *Semi-supervised Lyrics and Solo-Singing Alignment*. ISMIR, 2018.
35. **C. Gupta**, H. Li, and Y. Wang. *Perceptual Evaluation of Singing Quality*. APSIPA ASC, 2017. **Best Student Paper Award**.
36. D. Turnbull, **C. Gupta**, D. Murad, M. Barone, and Y. Wang. *Using Music Technology to Motivate Foreign Language Learning*. International Conference on Orange Technologies, 2017.
37. **C. Gupta**, D. Grunberg, P. Rao, and Y. Wang. *Towards Automatic Mispronunciation Detection in Singing*. ISMIR, 2017.
38. Z. Duan, **C. Gupta**, G. Percival, D. Grunberg, and Y. Wang. *SECCIMA: Singing and Ear Training for Children with Cochlear Implants via a Mobile Application*. SMC, 2017.
39. **C. Gupta** and P. Rao. *Objective Assessment of Ornamentation in Indian Classical Singing*. CMMR/FRSM 2011, Springer LNCS 7172, 2012.
40. V. Rao, **C. Gupta**, and P. Rao. *Context-aware Features for Singing Voice Detection in Polyphonic Music*. Adaptive Multimedia Retrieval, 2011.

Extended Abstracts, Workshop Papers, and Demonstrations

41. **C. Gupta**, N. V. Aritionang, V. Danry, T. Antoni, S. Nanayakkara, and P. Maes. *FactNudger: An AI-Powered Wearable Fact-Checker for Everyday Conversations*. ACM UIST, demonstration, 2026.
42. **C. Gupta**, A. Ram, S. Sridhar, C. Jouffrais, and S. Nanayakkara. *Scene-to-Audio: Distant Scene Sonification for Blind and Low Vision People*. ACM CHI Extended Abstracts, 2025.
43. M. H. Nguyen, M. A. Messerschmidt, **C. Gupta**, and S. Nanayakkara. *GutIO: Toward Sensing and Inducing Gut Feelings with Abdominal Sounds*. ACM CHI Extended Abstracts, 2025.
44. D. Dissanayake, **C. Gupta**, P. Sasikumar, and S. Nanayakkara. *VRsense: An Explainable System to Help Mitigate Cybersickness in VR Games*. ACM CHI Extended Abstracts, 2025.
45. M. Wu, P. Sasikumar, **C. Gupta**, S. Sridhar, P. L. Bagaskara, and S. Nanayakkara. *Factually: A Live, Wearable Fact-Checker*. ACM CHI Extended Abstracts, 2025.
46. **C. Gupta**, H. Wu, P. Sasikumar, S. Sridhar, P. L. Bagaskara, and S. Nanayakkara. *Factually: Exploring Wearable Fact-Checking for Augmented Truth Discernment*. ACM CHI Extended Abstracts, workshop paper, 2025.

Preprints and Manuscripts Under Review

1. **C. Gupta** et al. *VibeCheck: Proactive Wearable LLM Nudges for Spoken Misinformation In-the-Wild*. Under review for CHI 2027.
2. Latarache, **C. Gupta** et al. *Listening While Flying: AI-Assisted Acoustic Sensing for Drone-Based Search and Rescue*. Under review for CHI 2027.
3. Yadeesha, **C. Gupta** et al. *Problem-Solving or Instruction First? Exploring Instructional Sequencing for Second Language Vocabulary Learning with Wearable AI*. Under review for CHI 2027.

4. Vaidya, **C. Gupta** et al. *What Moves the Dial? Understanding Age and Task Characteristics Shaping Preferences for Computer-Use Voice Agent Autonomy*. Under review for CHI 2027.
5. **C. Gupta** et al. *DroneAudioNet: Noise Suppression for Drone Audition-based Search and Rescue*. Manuscript, 2026.